EPSR-test-ex39 Box-building chain: mixato, makelatticeato, randomise, fmole

Program(s): mixato, makelatticeato, randomise, fmole | chained passes: 1 | verification: ALL IDENTICAL (1 pass)

What this case tests

Runs the complete EPSR box-building workflow [1] in sequence — mix molecules, replicate the box, randomise positions and orientations, relax the molecules — as a user would before starting `epsr` [2,3]. With a Linux `system_commands.txt`, every file prompt goes through the real directory listing.

Code exercised

Fortran (original, fortran_src/):

- `mixato.f`: `mixato`; `makelatticeato.f`: `makelatticeato`
- `introtcluster.f`: `randomise`; `molfit.f`: `rotatm`, `molfit`
- `fmole.f`: `fmole`, `initpt_fmole`, `update_ato`; `ato_manipulation_rev4.f`: `fmolec1`, `generate_mp`
- `file_and_directory_commands.f`: `search_for_files` (file prompt / directory listing), `get_full_filename`; `ato_read_and_save.f`: `minitc` (read `.ato`), `moloutc` (write `.ato`)

Rust (port, rust/epsr-core/src/):

- `tools/mixato.rs`, `tools/makelatticeato.rs`, `tools/randomise.rs`, `tools/fmole.rs`: `run`
- `tools/setup_common.rs`: `search_for_files`, `SysCmds`, `rotatm`; `rng.rs`: `Ran1`
- `montecarlo.rs`: `molfit`; `energy.rs`: `fmolec1`; `potential.rs`: `generate_mp`; `ato.rs`: `minitc`, `moloutc`, `ftype`

Physical system

A dimethyl sulphoxide-water mixture [4] at 300 K: 4 DMSO + 8 water (64 atoms) at 0.1 atoms/Å³, cubic side 8.618 Å (`mixbox.ato`), replicated 2×2×2 to 96 molecules (32 DMSO + 64 water, 512 atoms) in a cubic box of 17.235 Å (`big.ato`), then randomised and shaken 10 times with the intramolecular Monte Carlo of `fmole` [5].

Inputs

`DMSO.ato`, `WaterMolecule.ato` from EPSR-test; `system_commands.txt` (`system_ls ls`, `system_termination /`). `mixato.stdin` and `fmole.stdin` answer each listing prompt with `e` and a typed name (independent of the `ls` sort order). Commands: `mixato` < `mixato.stdin`, `makelatticeato mixbox 2 2 2 big n`, `randomise big`, `fmole` < `fmole.stdin` (`big`, 10 shakes).

How the accuracy was verified

`commands.txt` (4 program calls, run once) was run with the Fortran reference build (`rust/tools/build_fortran_reference.sh`: the original sources, gfortran -O2 -fno-tree-vectorize, single thread) and with the Rust `epsr-rs` on identical copies of the inputs, and every file written was compared byte for byte (`rust/tools/run_ex_test.sh`, `compare_outputs.py`; no masking except where noted below). The Fortran output is stored in `fortran_reference.tar.gz`. Result of the final regression run of the complete port: every file identical in 1 pass.

Four program calls in one pass, each consuming the previous output, so the random-number state is carried through `randomise` and `fmole`. Compared outputs: `mixbox.ato`, `big.ato` and `dirlist.txt` (the listing `./DMSO.ato`, `./WaterMolecule.ato`, `./big.ato`, `./mixbox.ato`), all identical.

References

[1] A. K. Soper, "Empirical Potential Structure Refinement - EPSRshell: a user's guide", Technical Report RAL-TR-2011-012, STFC Rutherford Appleton Laboratory (2011).
[2] A. K. Soper, "Empirical potential Monte Carlo simulation of fluid structure", Chem. Phys. 202, 295-306 (1996).
[3] A. K. Soper, "Partial structure factors from disordered materials diffraction data: An approach using empirical potential structure refinement", Phys. Rev. B 72, 104204 (2005).
[4] A. K. Soper and A. Luzar, "A neutron diffraction study of dimethyl sulphoxide-water mixtures", J. Chem. Phys. 97, 1320-1331 (1992).
[5] N. Metropolis, A. W. Rosenbluth, M. N. Rosenbluth, A. H. Teller and E. Teller, "Equation of state calculations by fast computing machines", J. Chem. Phys. 21, 1087-1092 (1953).
[6] GNU Fortran (gfortran) 15.2, GCC, https://gcc.gnu.org/fortran/ - reference build with -O2 -fno-tree-vectorize.